Olist E-Commerce Sales Analytics

Dataset Overview

The Olist E-Commerce Sales Analytics project is an end-to-end data analytics solution developed to analyze sales performance, customer behavior, seller performance, and operational efficiency using the Olist Brazilian E-Commerce Public Dataset. The project transforms raw transactional data into meaningful business insights through data preprocessing, SQL analysis, and interactive dashboard visualization.

Methodology

This project followed an 8-phase workflow:

1. **Business Understanding** — defined business questions and success metrics

2. **Data Cleaning (Python)** — handled missing values, type conversions, duplicates

3. **Feature Engineering (Python)** — created delivery time, delivery delay, approval time metrics

4. **Exploratory Data Analysis (Python)** — visualized initial patterns

5. **SQL Analysis (MySQL)** — wrote business-question-driven queries, including RFM segmentation

6. **Power BI Dashboard** — built an interactive, single-page dashboard

7. **Business Insights** — translated findings into recommendations

8. **Documentation** — this report

Dataset Summary

The project uses the Olist Brazilian E-Commerce Public Dataset, a publicly available dataset containing transactional records from a Brazilian online marketplace. The dataset includes information related to customers, orders, products, sellers, payments, reviews, and geolocation. It enables comprehensive analysis of sales performance, customer behavior, payment trends, and logistics.

Dataset Name: Olist Brazilian E-Commerce Public Dataset

Source: Kaggle

Domain: E-Commerce / Retail

Analysis Period: September 2016 – October 2018

Number of Tables Used: 9

Approximate Number of Orders: 99,000+

Approximate Number of Customers: 99,000+

Approximate Number of Sellers: 3,000+

Approximate Number of Products: 33,000+

```python
print(orders.shape)
print(customers.shape)
print(sellers.shape)
print(products.shape)
print(order_items.shape)
print(order_payments.shape)
print(order_reviews.shape)
print(geolocation.shape)
print(category_translation.shape)
```

[4]

```
(99441, 8)
(99441, 5)
(3095, 4)
(32951, 9)
(112650, 7)
(103886, 5)
(99224, 7)
(1000163, 5)
(71, 2)
```

Data Cleaning & Preparation

Key Cleaning Steps

- Converted 8 datetime columns across 3 tables from string to proper datetime format

- Converted zip code columns from integer to string (preserving leading zeros)

- Labeled 610 products with missing category names as `"unknown"` rather than dropping them, since they represented non-trivial revenue

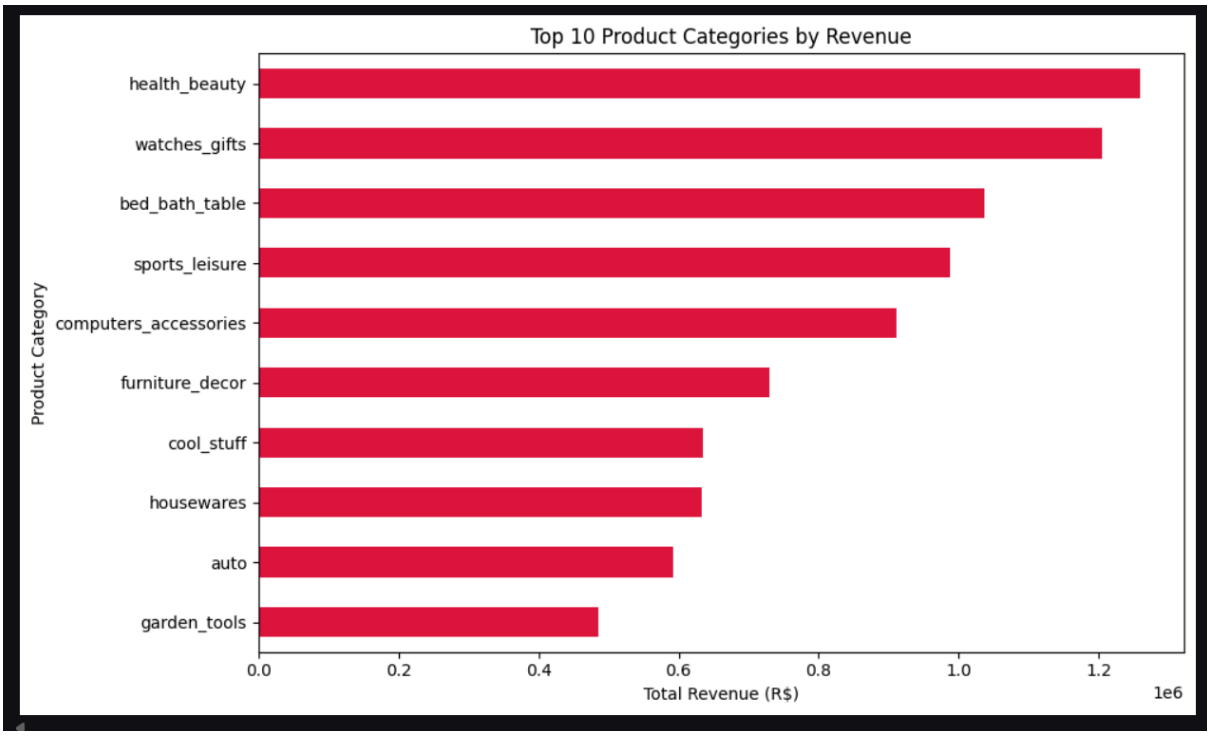
- Preserved missing delivery dates intentionally — these represent orders still in transit or lost, a meaningful lifecycle signal rather than a data error.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the overall characteristics of the dataset and identify important business patterns before building SQL queries and dashboards. Python visualization libraries were used to explore trends, distributions, and customer behavior.

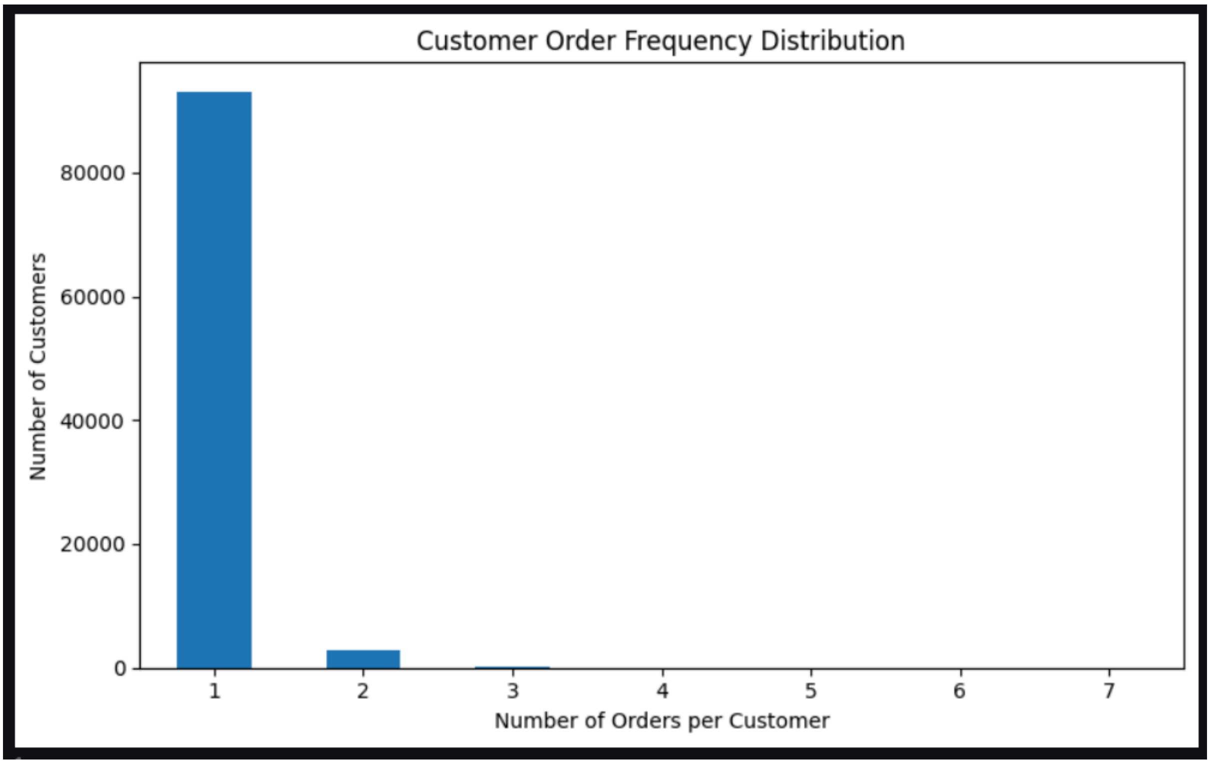
Revenue by Product Category

The analysis revealed that revenue is not evenly distributed across product categories. A few categories contribute significantly more revenue than others, indicating the importance of focusing marketing and inventory efforts on high-performing categories.



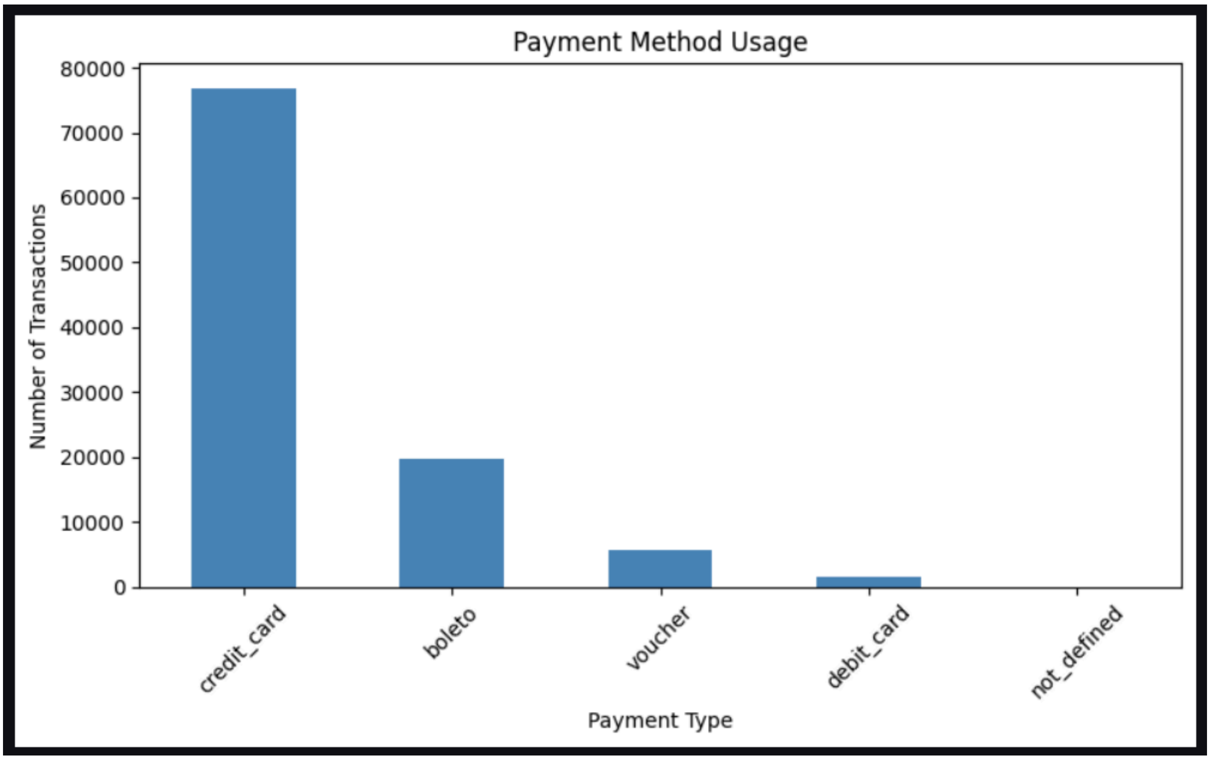
Customer Purchase Behaviour

Customer purchase analysis showed that the majority of customers placed only a single order, while repeat customers represented a much smaller proportion of the customer base. This highlights an opportunity to improve customer retention through loyalty programs and personalized marketing.



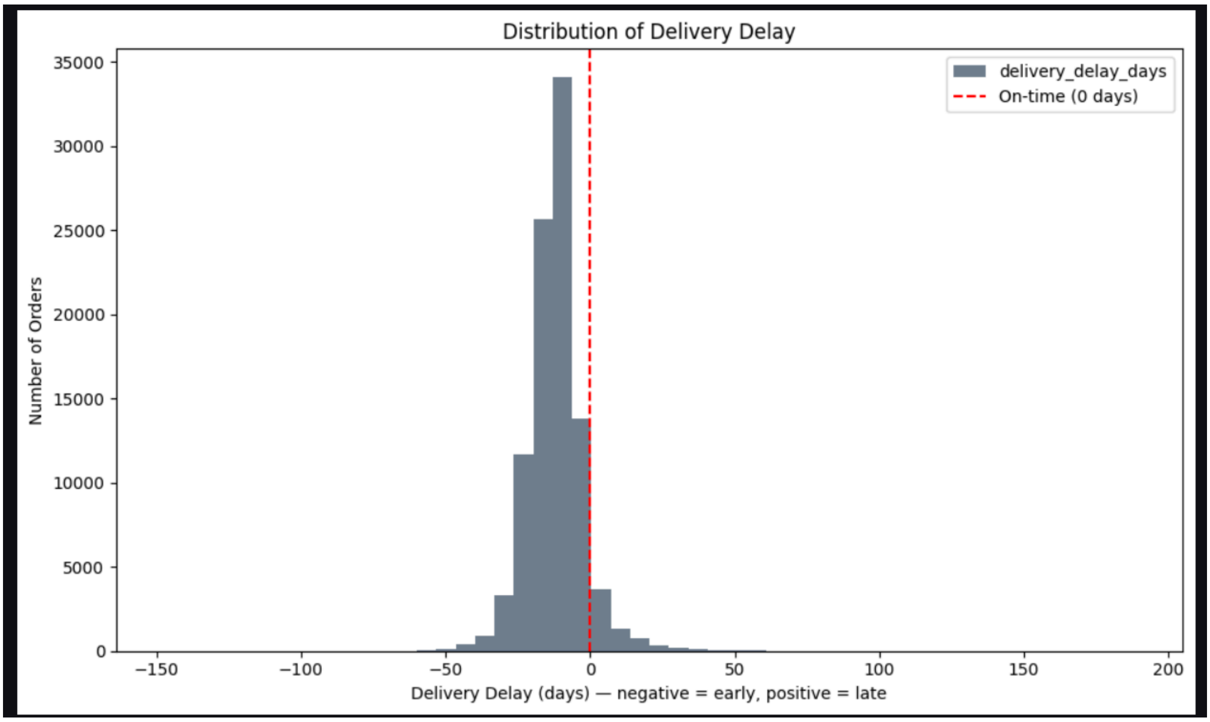
Payment Method Distribution

Different payment methods were analyzed to understand customer preferences. Credit card payments accounted for the largest share of transactions, making them the most preferred payment option among customers.



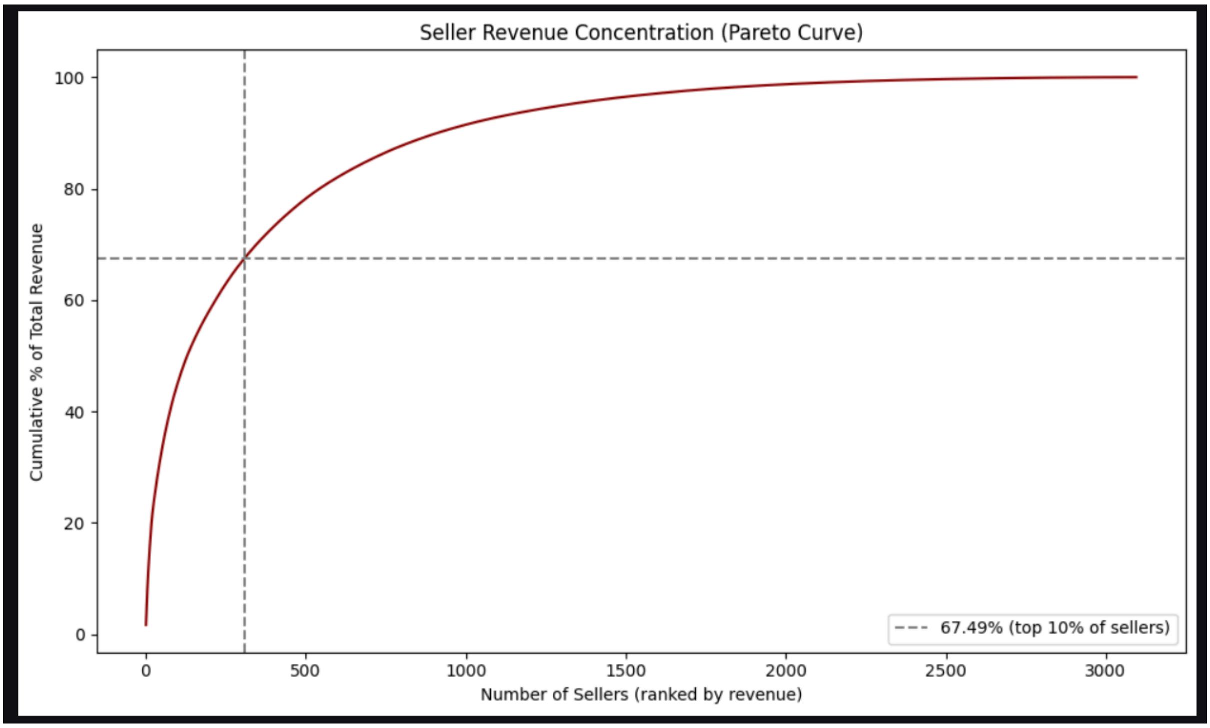
Delivery Performance

Delivery analysis was performed to understand shipping efficiency and identify delayed deliveries. Although many orders were delivered within the estimated timeframe, some orders experienced delays that may affect customer satisfaction.



Seller Contribution (Pareto Analysis)

The Pareto analysis demonstrated that a relatively small percentage of sellers generated a significant portion of total revenue. This follows the Pareto Principle (80/20 Rule), where a limited number of sellers contribute disproportionately to business performance.



SQL BUSINESS ANALYSIS

After preprocessing the data, the cleaned datasets were imported into MySQL. SQL was used to answer key business questions through joins, aggregations, grouping, filtering, and analytical queries. This phase transformed raw transactional data into meaningful business information.

Revenue Analysis

SQL queries were developed to calculate total revenue and identify the highest revenue-generating product categories.
Business Insight :
A limited number of product categories contributed a significant share of overall revenue, indicating strong sales concentration.

Customer Analysis

Customer purchasing behaviour was analyzed to identify customer segments and understand buying patterns.
Business Insight :
Most customers were one-time buyers, while loyal customers represented a smaller but highly valuable customer group.

Seller Performance

Seller performance was evaluated by calculating total revenue generated by each seller.
Business Insight :
A small number of sellers generated a large proportion of total marketplace revenue, confirming the Pareto distribution observed during EDA.

Payment Analysis

Payment data was analyzed to identify the most frequently used payment methods.
Business Insight :
Credit card payments dominated the marketplace, indicating customer preference for digital payment methods.

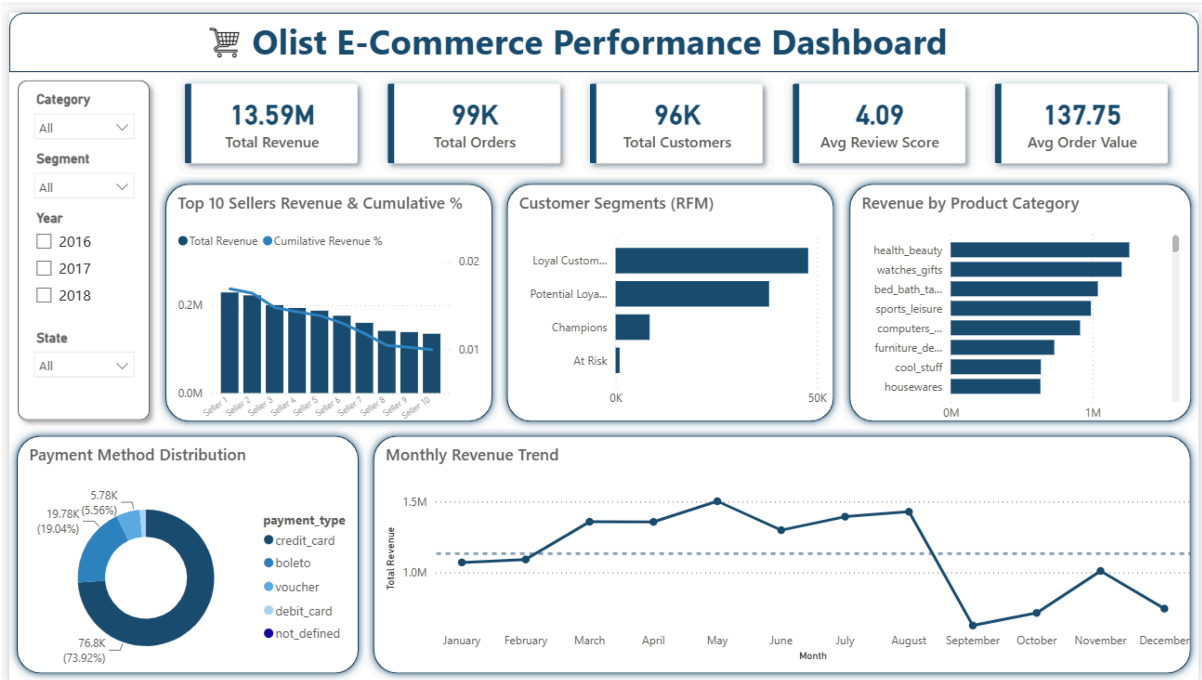
Delivery Performance Analysis

Delivery-related metrics were analyzed to evaluate logistics efficiency and identify delayed deliveries.
Business Insight :
Although the majority of deliveries were completed successfully, delayed deliveries highlighted opportunities to improve logistics operations.

POWER BI DASHBOARD

The final stage of the project involved developing an interactive dashboard using Microsoft Power BI. The dashboard was designed to provide business users with a centralized view of key performance indicators and enable interactive exploration of sales performance, customer behaviour, payment trends, and seller performance.

The dashboard follows a clean and user-friendly layout, allowing stakeholders to quickly identify important business trends and make informed decisions.



Dashboard Components:

- 5 KPI cards: Total Revenue, Total Orders, Total Customers, Avg Review Score, Average Order Value

- Customer Segments (RFM) — bar chart

- Payment Method Distribution — donut chart

- Revenue by Product Category — bar chart

- Top 10 Sellers: Revenue & Cumulative % — Pareto combo chart

- Total Revenue by Month — trend line

- Interactive filters: Category, Customer Segment, Year, State

Key Business Insights

The analysis of the Olist E-Commerce dataset produced several valuable business insights.

1. Product Category Performance

Health & Beauty emerged as one of the highest revenue-generating product categories, indicating strong customer demand.

2. Customer Behaviour

Most customers placed only a single order, suggesting opportunities to improve customer retention through loyalty programs and personalized marketing.

3. Payment Preferences

Credit Card was the most preferred payment method, accounting for the largest share of transactions.

4. Seller Performance

A relatively small percentage of sellers generated a significant portion of total marketplace revenue, following the Pareto Principle.

5. Revenue Trend

Monthly revenue fluctuated over the analysis period, highlighting seasonal demand patterns and changing customer purchasing behaviour.

6. Delivery Performance

Most deliveries were completed within the expected timeframe, while delayed deliveries indicated areas where logistics performance could be improved.

7. Customer Segmentation

RFM analysis identified different customer groups, enabling targeted marketing strategies based on purchasing behaviour.

8. Business Intelligence

The interactive dashboard provides stakeholders with a centralized platform to monitor KPIs, explore trends, and support strategic business decisions.

Business Recommendations

Based on the analysis, the following recommendations are proposed to improve business performance.

- Improve Customer Retention
  Introduce loyalty programs, personalized promotions, and targeted marketing campaigns to encourage repeat purchases.

- Focus on High-Performing Categories
  Allocate additional marketing resources and inventory to high-revenue product categories to maximize sales.

- Support Top Sellers
  Develop incentive programs for high-performing sellers while providing training and support to underperforming sellers.

- Optimize Delivery Performance
  Monitor delayed deliveries and collaborate with logistics partners to reduce delivery times and improve customer satisfaction.

- Enhance Payment Experience
  Continue supporting digital payment methods while encouraging customers to adopt secure online payment options.

- Data-Driven Decision Making
  Use interactive dashboards regularly to monitor business performance and identify emerging trends before making strategic decisions.

Folder Structure

```
Ecommerce_Sales_Analytics/
├── data/
│   ├── raw/
│   └── cleaned/
├── notebooks/
│   ├── 01_data_loading.ipynb
│   ├── 02_data_cleaning.ipynb
│   ├── 03_feature_engineering.ipynb
│   └── 04_eda.ipynb
├── sql/
│   ├── database_setup.sql
│   └── business_analysis.sql
├── powerbi/
│   └── Olist_Ecommerce_Analytics.pbix
├── images/
├── reports/
└── README.md
```